

Udemy ML, DL, NLP
course

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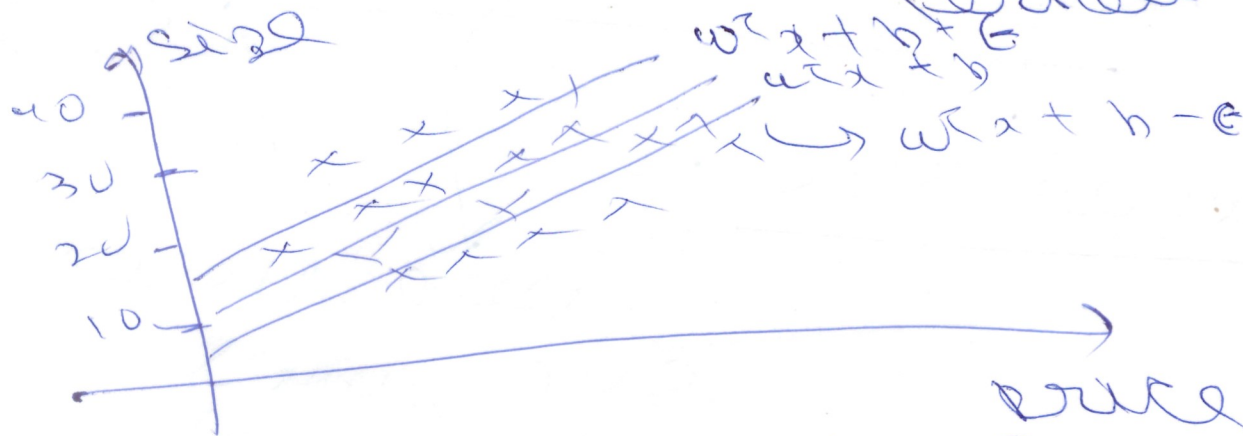
① Sec-36

Support Vector
machines

lec-5 Support
Vector
Regression

lec-6

~~Support~~ SVM
kernels



ϵ :- marginal error

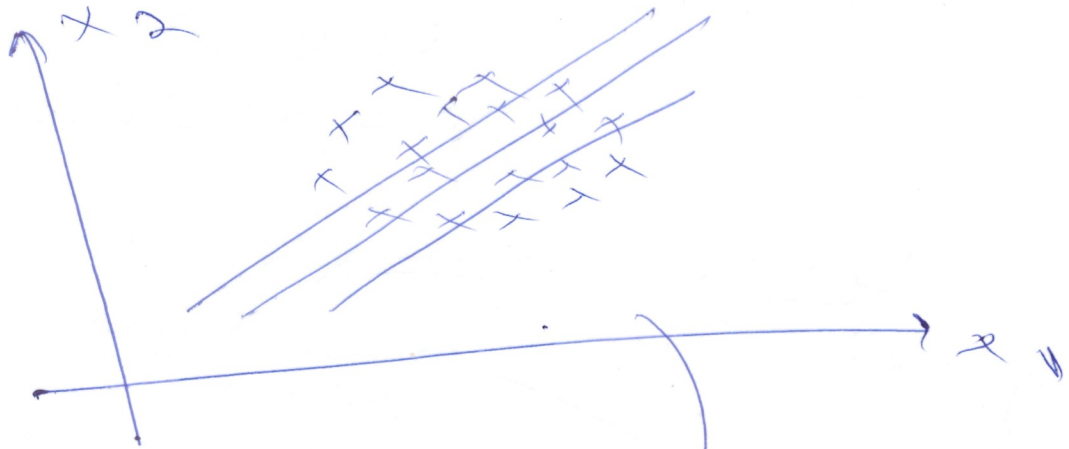
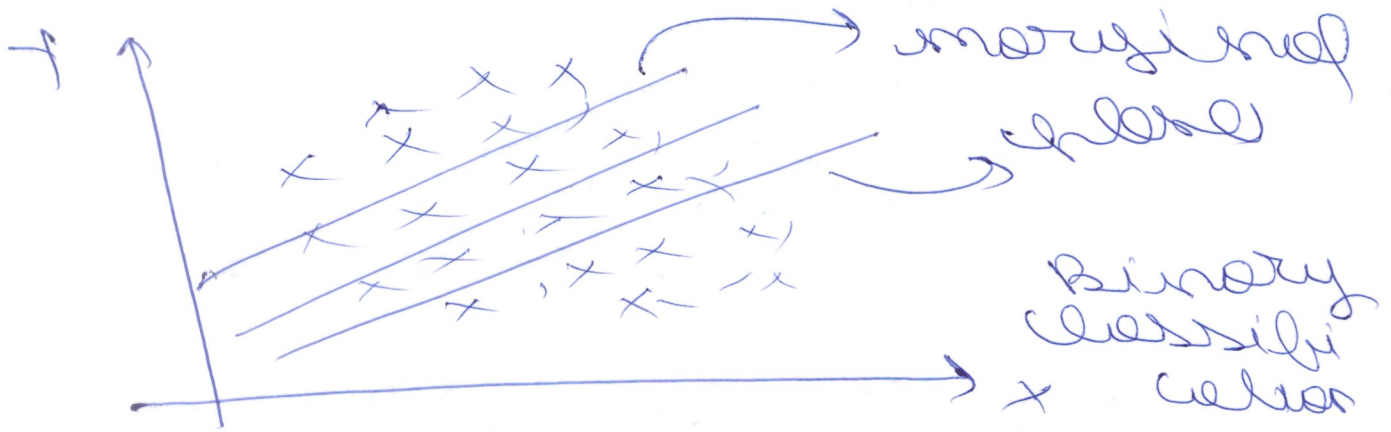
min $\frac{\|w\|^2}{2} + \left(\sum_{i=1}^n \xi_i \right)$

Consistent :-

$$(y = w(x)) \leq \epsilon + \delta$$

$\Downarrow$
loss function

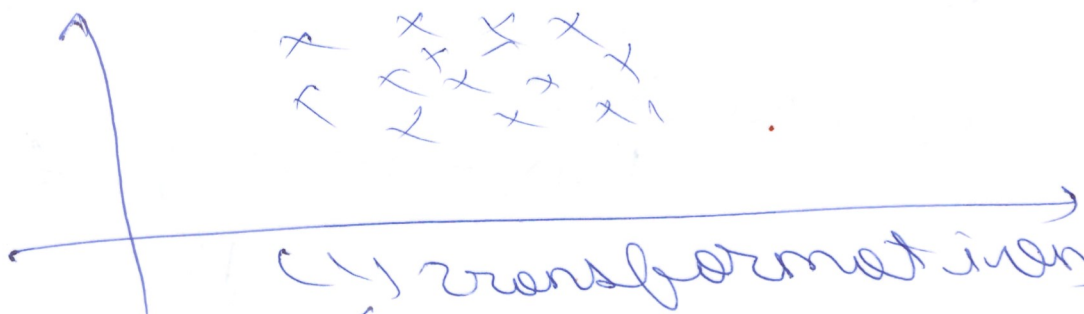
⇒ SVM Kernel



Linear SVC :-

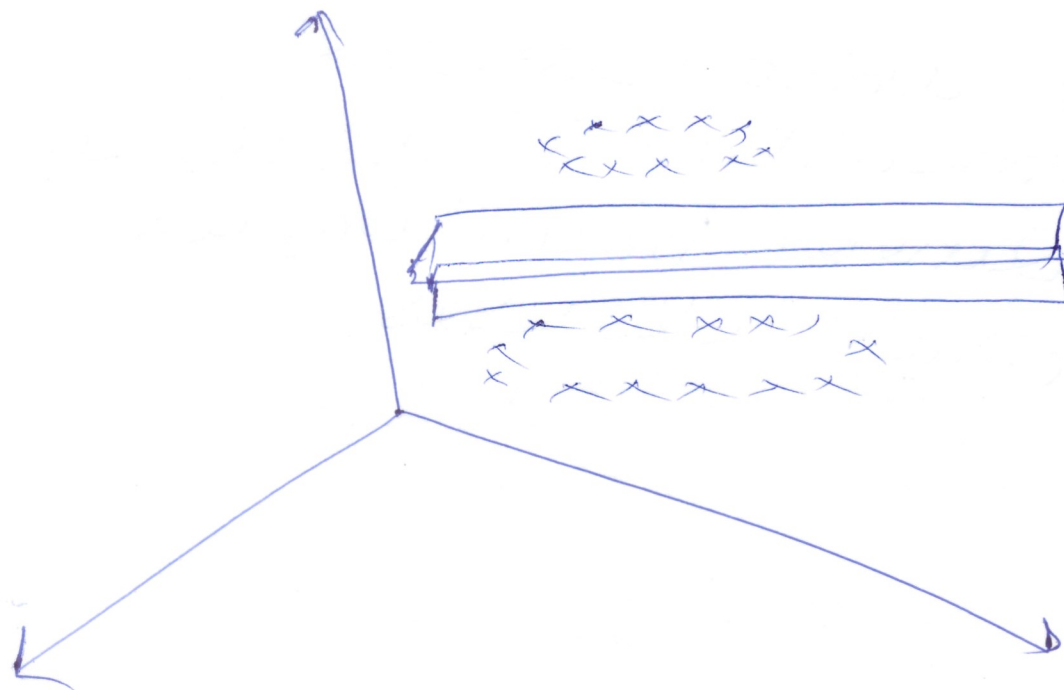
linear over

SVC work as all points are overlapped.

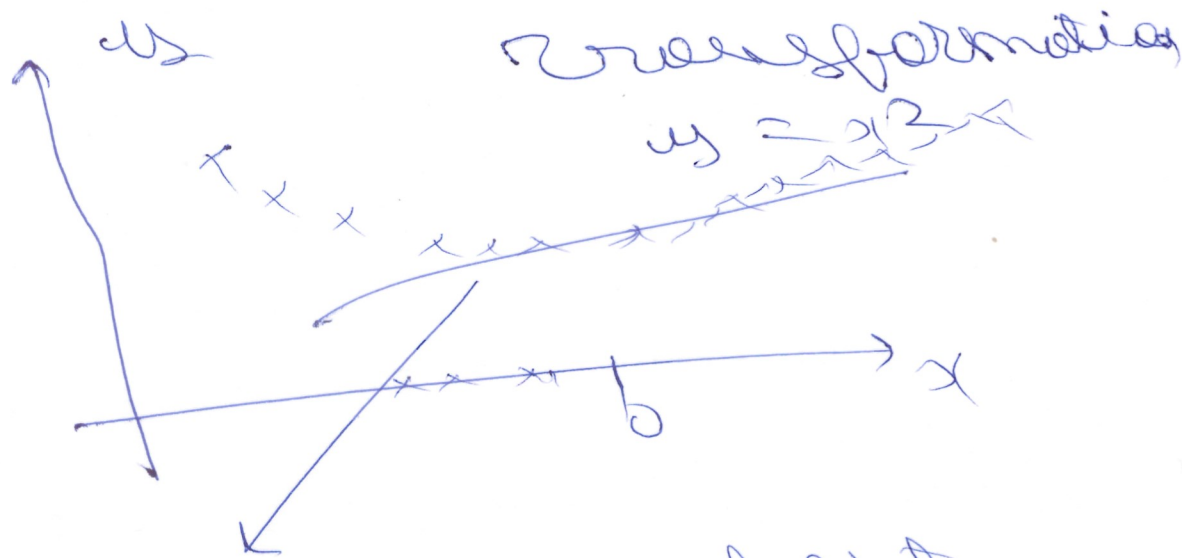


mathematical transformation

3



SVC to be applied



crossed a line by

Linear SVC

Basically
kernel in SVC is a mathe-
matical function that trans-
forms non linearly separable
data into linearly separable

4
split into higher
dimensional space where
a linear boundary can
separate classes.